

Chapter 32

Electromagnetic Waves



Maxwell's Equations

Before Maxwell:

- Gauss' law:

$$\oiint \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} Q_{enc}$$

- Gauss' law for magnetism:

$$\oiint \vec{B} \cdot d\vec{a} = 0$$

- Ampere's Law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$$

- Faraday's Law:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \iint \vec{B} \cdot d\vec{a}$$

Maxwell's Equations

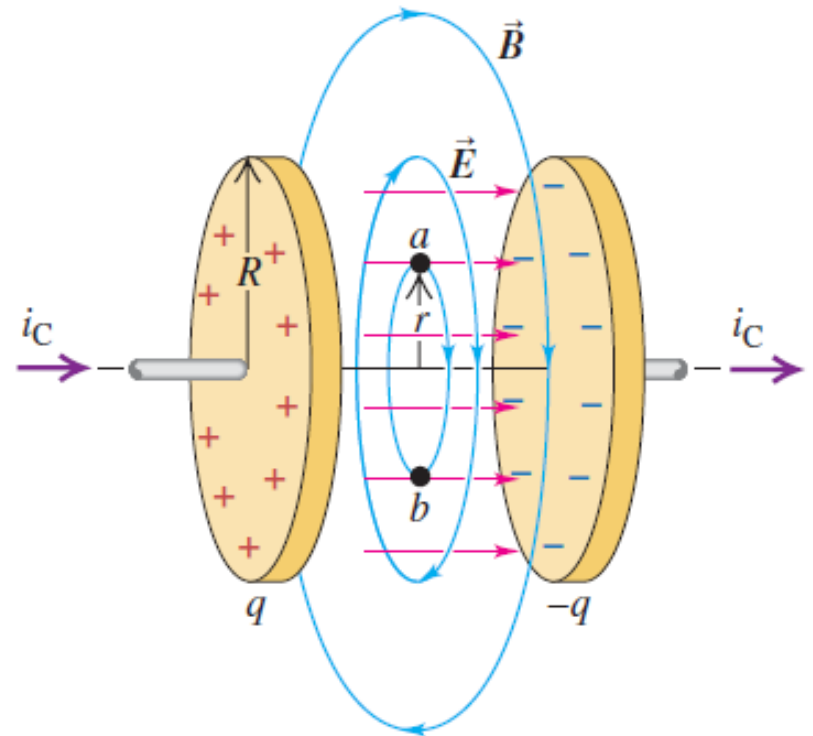
Maxwell's fix - Modified Ampere's law:

- On charging a capacitor, there is no current between capacitor plates, but we have a *changing electric field*.

$$E = \frac{Q}{\epsilon_0 A} \Rightarrow Q = \epsilon_0 A E = \epsilon_0 \Phi_e$$

$$\frac{dQ}{dt} = \epsilon_0 \frac{d\Phi_e}{dt} = I_d$$

The displacement current



Maxwell's Equations

Maxwell's fix - Modified Ampere's law:

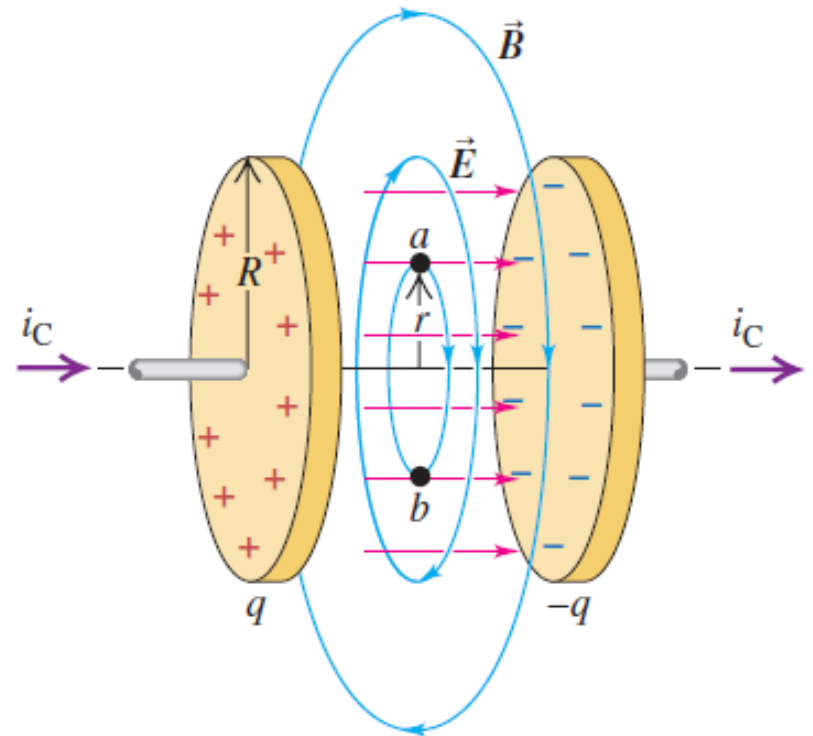
- To modify Ampere's law

$$I_d = \epsilon_0 \frac{d\Phi_e}{dt}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0(I_{enc} + I_d)$$

$$= \mu_0 I_{enc} + \mu_0 \epsilon_0 \frac{d\Phi_e}{dt}$$

$$= \mu_0 I_{enc} + \mu_0 \epsilon_0 \frac{d}{dt} \iint \vec{E} \cdot d\vec{a}$$



Maxwell's Equations

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$$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \iint \vec{B} \cdot d\vec{a}$$

Maxwell's Equations

Maxwell's equations:

- Combining Maxwell's equations in empty space results in:

$$\frac{\partial^2 \mathbf{E}}{\partial x^2} = \epsilon_0 \mu_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \text{and} \quad \frac{\partial^2 \mathbf{B}}{\partial x^2} = \epsilon_0 \mu_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}$$

Recall the wave equation:

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

- We identify the wave speed

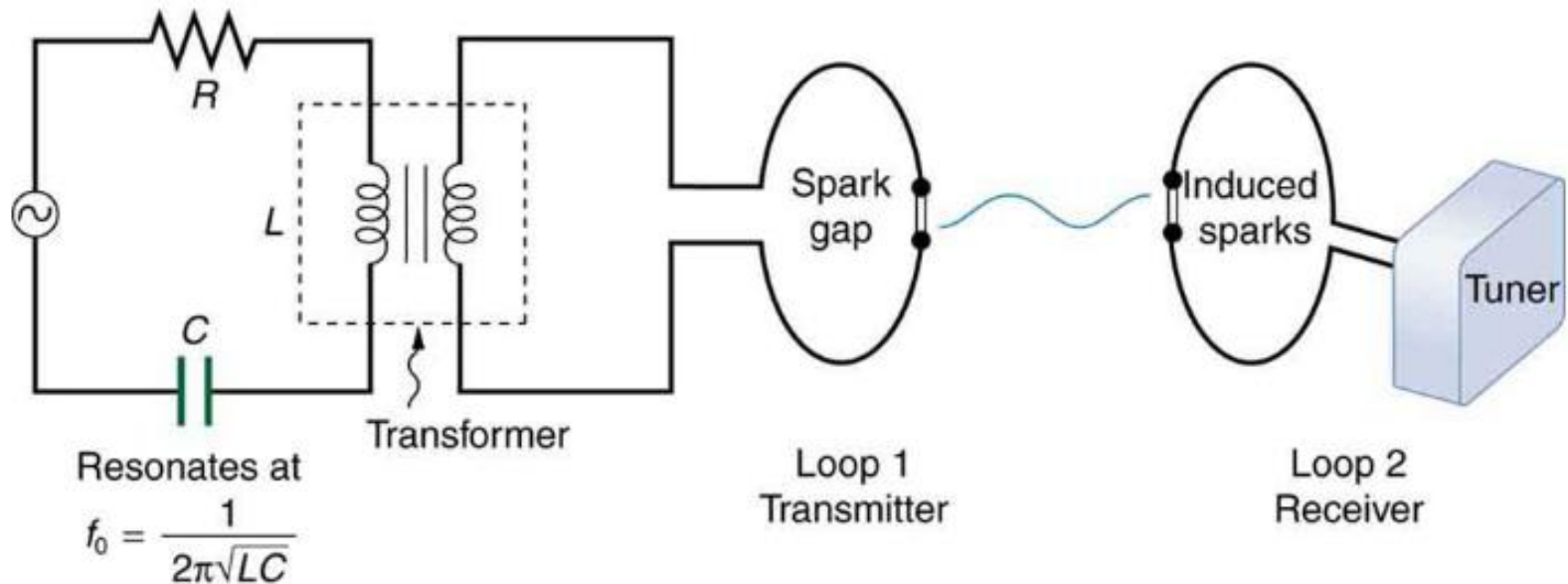
$$\frac{1}{v^2} = \epsilon_0 \mu_0 \quad \text{or} \quad v = \frac{1}{\sqrt{\epsilon_0 \mu_0}} = 3 \times 10^8 \text{ m/s}$$

Maxwell's Equations

Electromagnetic waves: Hertz's experiment

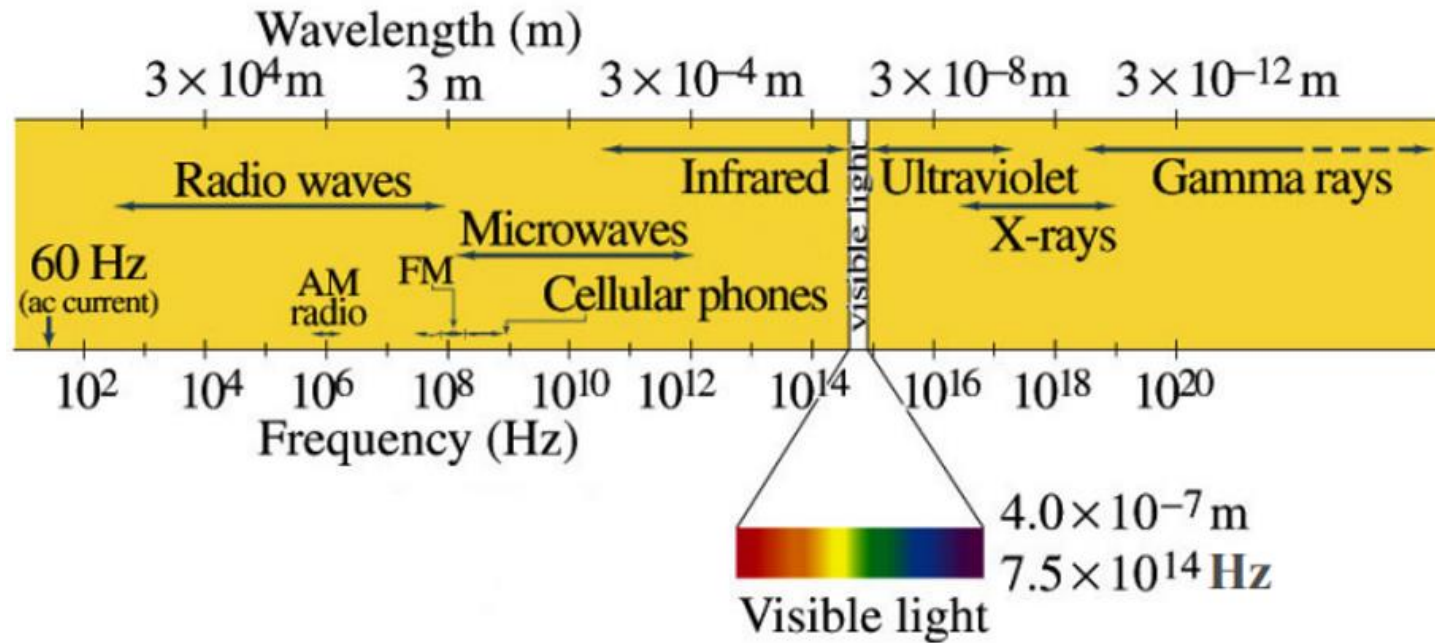
Apparatus used by Hertz in 1887

- An RLC circuit connected to the first loop caused sparks across a gap in the wire loop and generated electro-magnetic waves.
- Sparks across a gap in the second loop located across the laboratory detected the waves.



Maxwell's Equations

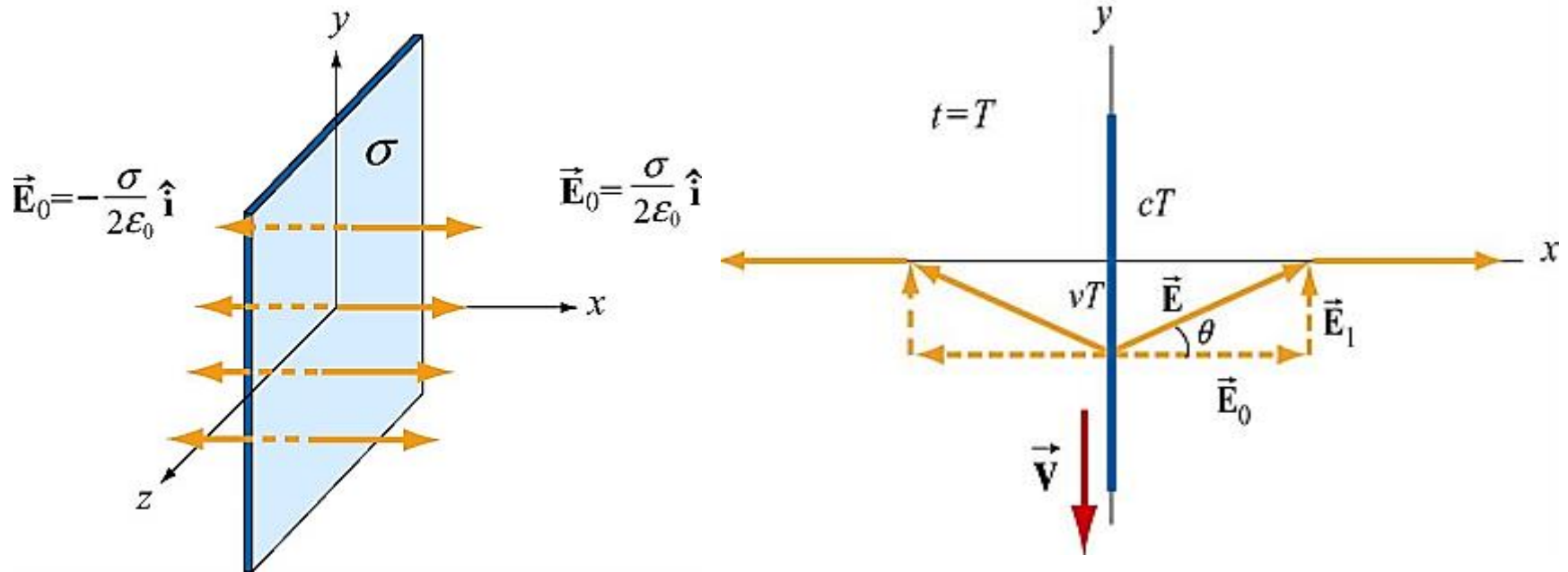
Electromagnetic spectrum:



Plane Electromagnetic Waves

Generating plane waves:

- Shake a charged sheet up and down:

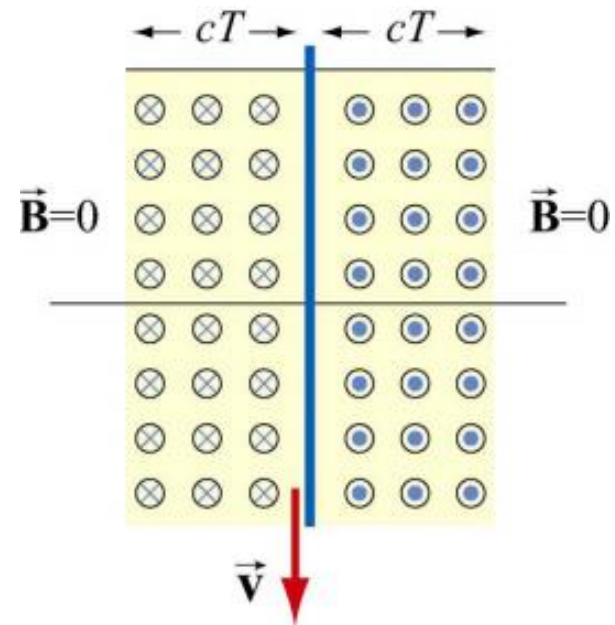
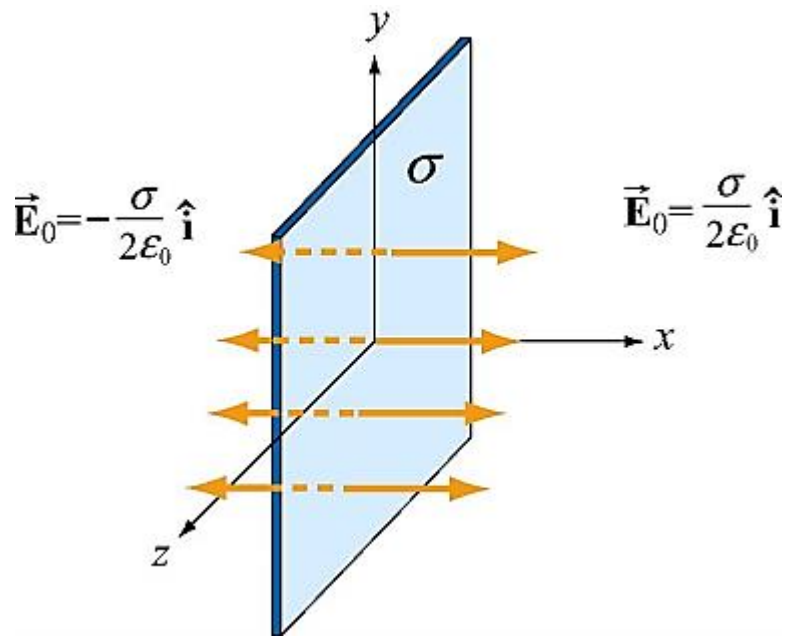


A transverse component $\vec{E}_1$ of the electric field $\vec{E}$ arises.

Plane Electromagnetic Waves

Generating plane waves:

- Shake a charged sheet up and down:

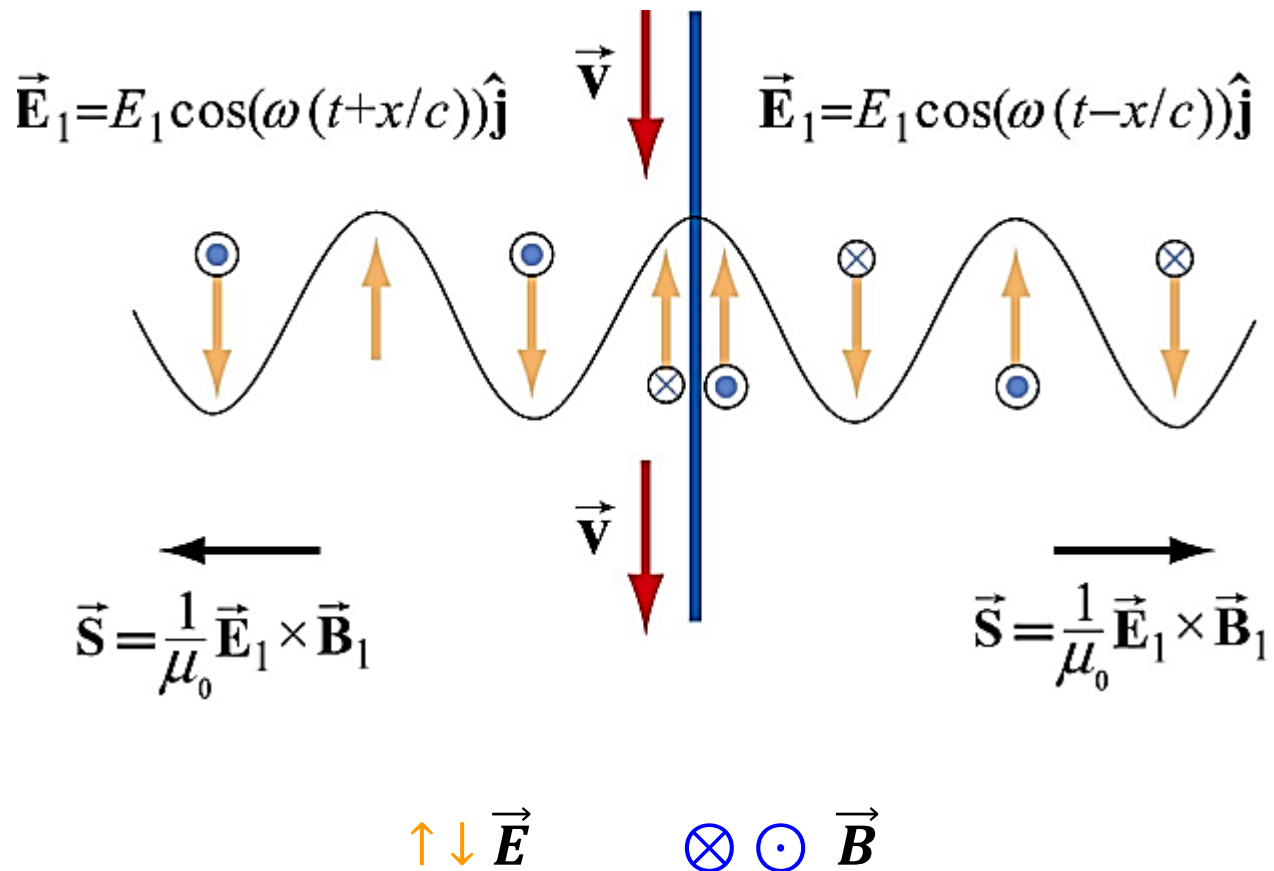


The induced magnetic field $\vec{B}$.

Sinusoidal Electromagnetic Waves

Generating plane harmonic waves:

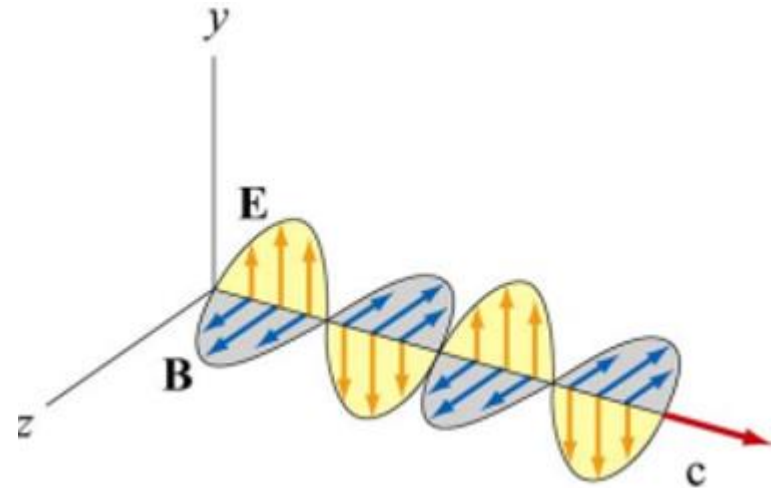
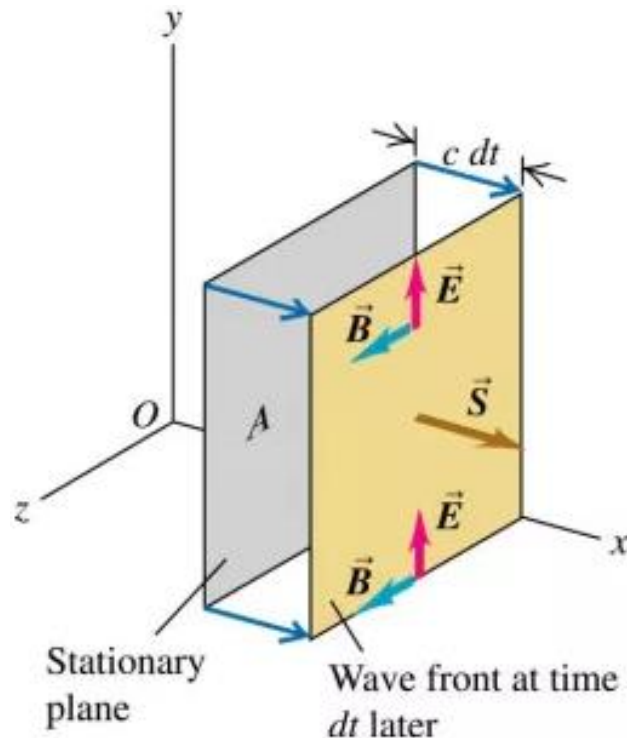
- Shake a charged sheet up and down sinusoidally :



Sinusoidal Electromagnetic Waves

Generating plane harmonic waves:

- Shake a charged sheet up and down sinusoidally :*



$$E_y = E_{max} \cos(kx - \omega t) \quad \text{and} \quad B_z = B_{max} \cos(kx - \omega t)$$

Energy and Momentum in Electromagnetic Waves

The Poynting vector:

- The fields $\vec{E}$ and $\vec{B}$ are mutually perpendicular with:

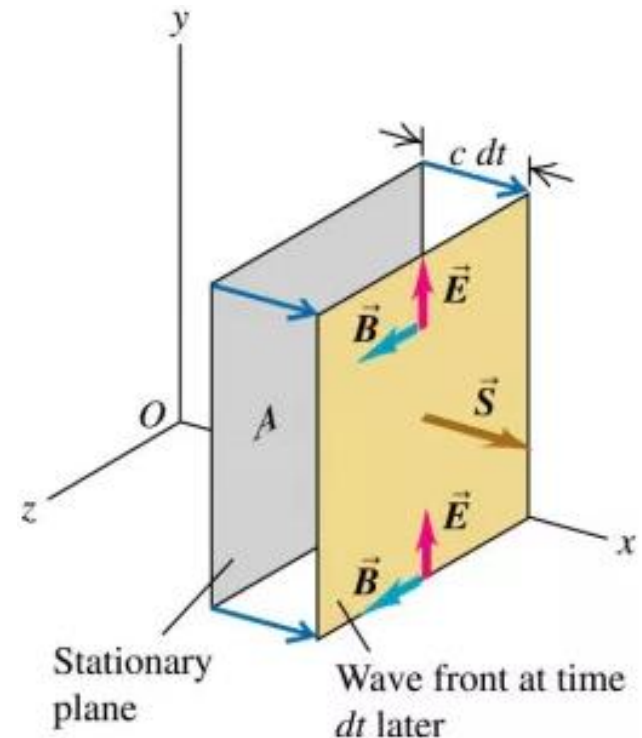
$$B = \epsilon_0 \mu_0 c E \quad \text{or} \quad E = cB$$

$(\vec{E} \times \vec{B})$ points in direction of x

- Define the Poynting vector $\vec{S}$:

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

$$|\vec{S}| = \frac{EB}{\mu_0} = \frac{E^2}{c\mu_0}$$



Energy and Momentum in Electromagnetic Waves

The Poynting vector:

- The total energy density:*

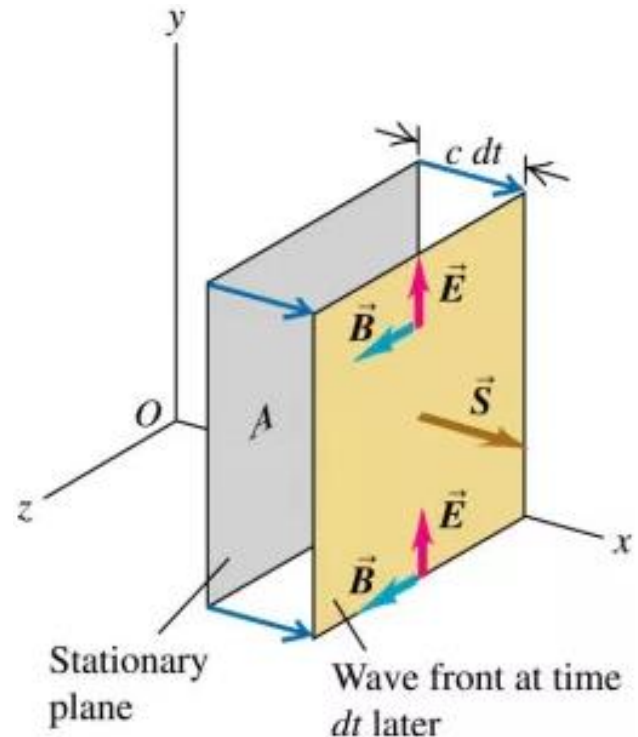
$$u = \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2\mu_0} B^2$$
$$= \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2\mu_0} \left(\frac{E}{c}\right)^2 = \epsilon_0 E^2$$

- Energy propagates in space :*

$$dU = u dV = (\epsilon_0 E^2) [(A)(c dt)]$$

$$\frac{1}{A} \frac{dU}{dt} = c\epsilon_0 E^2 = \frac{E^2}{c\mu_0} = \frac{EB}{\mu_0} = |\vec{S}|$$

Intensity of radiation (Units: Watts/m²)



Energy and Momentum in Electromagnetic Waves

The Poynting vector:

- Intensity:

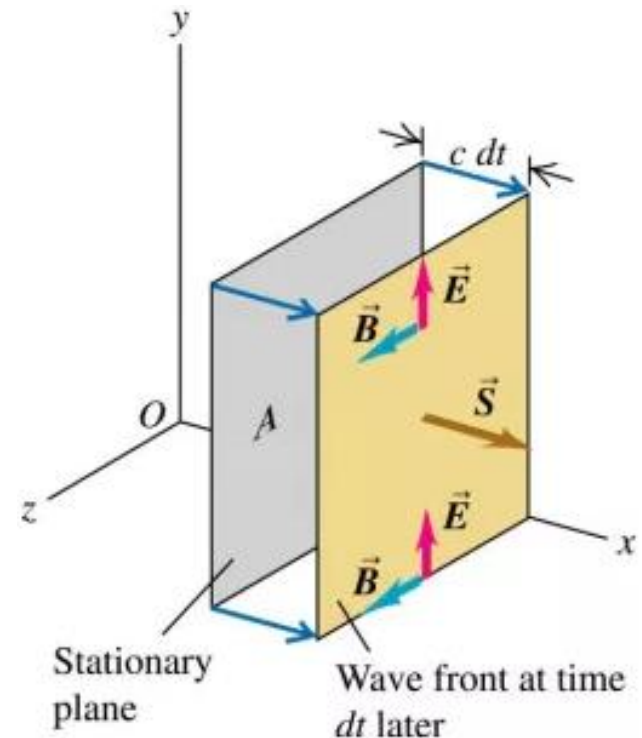
$$E_y = E_0 \cos(kx - \omega t)$$

$$B_z = B_0 \cos(kx - \omega t)$$

$$|\vec{S}| = \frac{E_0 B_0}{\mu_0} (\cos)^2(kx - \omega t)$$

- The average intensity:

$$I = |\vec{S}|_{ave} = \frac{E_0 B_0}{2\mu_0} = \frac{E_0^2}{2c\mu_0} = \frac{cB_0^2}{2\mu_0}$$



Energy and Momentum in Electromagnetic Waves

The Poynting vector:

- *EM waves also carry momentum:*

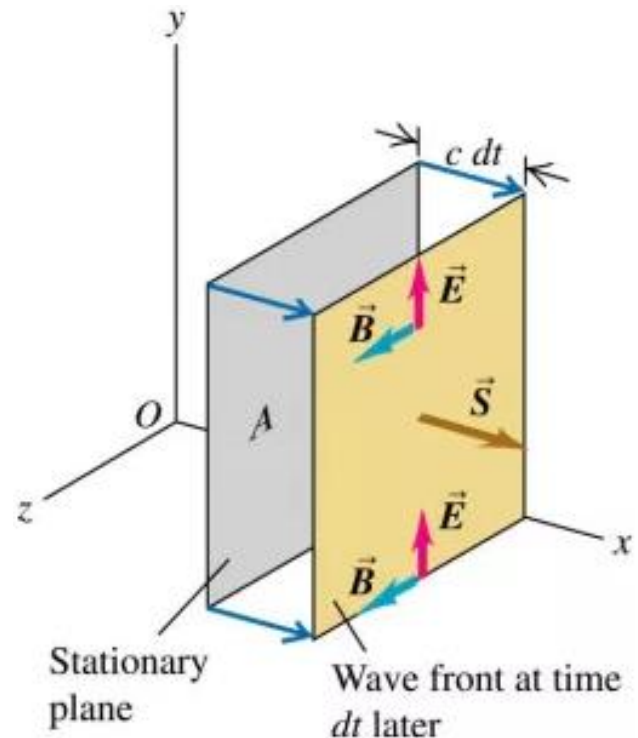
$$\text{Momentum density} = \frac{d|\vec{p}|}{dV}$$

$$= \frac{EB}{\mu_0 c^2} = \frac{|\vec{S}|}{c^2}$$

- *Momentum flow rate:*

$$\frac{d|\vec{p}|}{Adt} = \frac{|\vec{S}|}{c} = \frac{EB}{c\mu_0}$$

Responsible for radiation pressure



END

